Delphes vs. CMS for HH→ $b\bar{b}\tau\tau$:
where we stand, and one decision to take

August 12, 2026

Abstract

This note has three parts. **Section 1** audits what the Delphes fast simulation currently does relative to CMS NanoAOD, object by object: what is matched, what is tuned to a CMS anchor, what differences remain by choice or by limitation, and which of the ten NSBI input features still disagree. **Section 2** states the one open decision — whether to continue correcting the jet-based τ_h or to rebuild it from the Delphes particle-flow candidates — with the considerations on each side and the questions we would like τ expertise on. **Section 3** is a reference list of the studies behind the numbers, for anyone who wants to check a specific claim.

Short summary: the $\tau_h\tau_h$ yield deficit that motivated this work (a factor ∼5) is closed, the cutflow now agrees with CMS at every stage (1.01), and eight of the ten NSBI features agree within statistics. The two that do not are $m_{\tau\tau}$ (residual ∼7 GeV offset) and $\Delta R_{\tau\tau}$ (an excess at 2.2–3.2). Since the previous version the $m_{\tau\tau}$ offset has been *localised*: it lives in the τ four-vectors, not in the mass estimator or the p_T^{miss}, and a specific mechanism in the energy-scale application is identified with one test outstanding (Section 1.4). Two other candidates — the sampled τ_h mass and a τ energy resolution mismatch — were measured and excluded at the 0.1% level.

1 Audit: what we do, relative to CMS

The Delphes card is the stock CMS card plus the patches in `docs/card_patch_validation.md`; all object tuning is applied *downstream*, so the produced samples are never re-made. Every efficiency and scale is measured on a private 2024 NanoAODv15 anchor and applied to Delphes, so "tuned" below always means "measured on CMS and transferred".

1.1 Matched or tuned to CMS

Table 1 lists the quantities that either agree with CMS already or are transferred from it by a measured map. In every case the map is derived on the anchor and applied to Delphes downstream; none of them requires a change to the card.

1.2 Residual differences, by choice or by limitation

1. **A τ_h is an AK4 jet.** This is the structural difference. In CMS a τ_h is an HPS object built from a narrow set of decay products; in Delphes it is an $R = 0.4$ jet carrying `TauTag`. Its mass, energy and direction are therefore all wrong at source, and we correct the first two against the anchor rather than fixing the representation. This is the subject of Section 2.

2. **The τ_h fake rate is measured on signal.** `tau_mistag` comes from the b-rich HH→ $b\bar{b}\tau\tau$ anchor. Fake rates are strongly jet-flavour dependent, so this value is not appropriate for the fake-dominated backgrounds (QCD, W+jets); those need their own measurement before they are trusted.

3. p_T^{miss} **smearing is isotropic and flat.** Real p_T^{miss} resolution is anisotropic (larger along the hadronic recoil) and grows with ΣE_T. Ours is round and constant, which reproduces the overall width but not its structure — and $m_{\tau\tau}$ is now marginally *broader* on Delphes than on CMS, as one would expect from over-simplified noise.

quantity	status	how
b-tag efficiency (b, c, light)	✓ tuned	stochastic re-tag from `Jet.Flavor` at the anchor-measured UParT-AK4 Medium efficiency
b-jet energy scale	✓ tuned	response inverted to unity against `GenJet`
m_{bb} peak and width	✓ matched	follows from the above
τ_h identification efficiency	✓ tuned	anchor `GenVisTau`→DeepTau v2.5 Medium; includes CMS's HPS reconstruction inefficiency
τ_h visible mass	✓ tuned	*sampled* per jet from the anchor's per-p_T mass quantiles
τ_h energy scale	(×) tuned	CMS/Delphes response ratio, both against a *visible* τ reference — but it under-corrects; see Section 1.4
τ_h acceptance	✓ identical	common $p_T > 20\,\text{GeV}$, $\lvert\eta\rvert < 2.3$ on both sides
jet–τ overlap removal	✓ identical	di-τ pair chosen first, then $\Delta R > 0.4$ on both sides
p_T^{miss} resolution	✓ tuned	Gaussian smearing to the anchor resolution ($16.5 \to 32.6\,\text{GeV}$ per component)
lepton reconstruction efficiency	✓ tuned	per-p_T scale factors (large: $1.57 \to 1.22$ for e, $1.30 \to 1.13$ for μ)
di-τ mass estimator	✓ identical	the same covariance-free FastMTT is run on both sides

Table 1: Quantities that agree with CMS or are transferred from it.

4. **Pileup enters only through p_T^{miss}.** The card runs without pileup by design. The jet energy *resolution* is therefore untuned (only the scale is corrected), and pileup jets cannot be produced at all. Both act on the $b\bar{b}$ side, which currently agrees well; the second cannot be fixed downstream in any case.

5. **The b-jet energy scale is self-anchored.** It is corrected to its own `GenJet` rather than to CMS. This is defensible — CMS jet energies are JEC-calibrated to gen — and m_{bb} agrees, but it is not the like-for-like comparison used everywhere else.

6. **The sampled τ_h mass is drawn independently** of the τ energy and of the partner leg, whereas the true visible mass correlates with the decay mode and hence with prong multiplicity and energy sharing.

7. **Fake τ_h receive no energy correction.** The energy-scale factor is selected per jet by gen-matching: `bjet_escale` for `flavor==5`, `tau_escale` for jets within $\Delta R < 0.4$ of a hadronic gen τ (τ precedence), and 1 otherwise. A *fake* τ_h — a light jet promoted by the stochastic tag draw — satisfies neither condition and is therefore left uncorrected, while still receiving a τ-like visible mass. A related, smaller mismatch: the match is made against *all* gen τ (`|pid|==15`), including leptonically decaying ones, whereas the map is derived on the hadronic visible set only.

8. **FastMTT is covariance-free** (a fixed p_T^{miss} resolution rather than the per-event covariance). This is run identically on both sides, so it largely cancels in the comparison; it limits the absolute $m_{\tau\tau}$ resolution, not the agreement.

1.3 The ten NSBI features

What is known about the two that disagree. A dedicated study (Section 3, item 6) compared the *generator-level* $\Delta R_{\tau\tau}$ and the selection efficiency against it, on both sides. The gen

feature	status	comment
m_{HH}	✓	the κ_λ-carrying observable; agrees across $\kappa_\lambda = 0, 1, 5$
m_{bb}	✓	both peak near 112 GeV
p_T^{HH}	✓	
ΔR_{bb}	✓	after symmetric overlap removal on both sides
$\Delta\phi_{HH}$	✓	
p_T^{H1}, p_T^{H2}	✓	
$\cos\theta^*$	✓	statistically limited but consistent
$m_{\tau\tau}$	(×)	residual ∼7 GeV high; localised to the τ energy scale, Section 1.4
$\Delta R_{\tau\tau}$	×	Delphes excess at 2.2–3.2

Table 2: Agreement of the NSBI inputs with CMS at $\kappa_\lambda = 0$ (the other κ_λ points behave the same). Yields: gen-matched cutflow 1.01; analysis-level selection 1.18×.

spectra agree — so the samples are the same physics and no generator-level explanation is available — but the efficiencies differ: Delphes selects 0.22–0.23 of events at gen $\Delta R > 2.5$ where CMS selects 0.15–0.17 ($\sim 3\sigma$), with a weaker ($\sim 2\sigma$) hint that Delphes also loses pairs below $\Delta R < 1$. Both push the normalised distribution rightward. Three explanations were proposed and excluded by measurement: fake τ_h (fake fractions are 5.7% Delphes against 5.0% CMS, and the discrepancy lives entirely in the gen-matched component); τ-pair selection (making it fully symmetric moved $\Delta R_{\tau\tau}$ but not $m_{\tau\tau}$); and a large CMS τ energy miscalibration (the anchor response is 0.97–0.99).

1.4 $m_{\tau\tau}$: where the offset lives

Because $m_{\tau\tau} = m_{\text{vis}}/\sqrt{x_1 x_2}$ factorises, the *visible* di-τ mass m_{vis} — built from the two selected τ four-vectors, with no p_T^{miss} and no FastMTT — separates a τ energy-scale error from an error in the x fit. The two have disjoint fixes, so it is worth measuring before tuning anything. On the $\kappa_\lambda = 0$ overlay m_{vis} is offset by *at least* as much as $m_{\tau\tau}$ is (peak ratios ≈ 1.24 against ≈ 1.17, read off the histograms), so the τ four-vectors carry the discrepancy and the FastMTT fit is not its source — if anything it compresses it slightly. The individual leg spectra agree: both τ legs are harder on Delphes.

Two features of that panel are not explained by a pure scale. Delphes populates $m_{\text{vis}} > 125$ GeV, which is kinematically impossible for $H \to \tau\tau$ — a visible mass cannot exceed its parent — and CMS respects that edge; and scaling the CMS distribution by 1.24 reaches ~ 155 GeV, not the ~ 200 GeV observed. The uncorrected fakes of Section 1.2 would produce both: the offset, because the correction is measured only on the gen-matched population, which least needs it; and the tail, because a fake τ_h carries a full AK4 jet p_T with no kinematic ceiling from m_H. Consistently, with an applied scale of $\simeq 0.85$ and a residual ratio of $\simeq 1.24$, the raw Delphes τ are $\sim 46\%$ harder than CMS while the map corrects only 15%.

This is **not yet confirmed**. The measured fake fractions (5.7% Delphes, 5.0% CMS) look too small to account for the whole excess, so the pending test is the gen-matched/fake split of m_{vis}, which the existing tooling already produces. If the excess turns out to be largely *gen-matched*, the fake mechanism is not the explanation and the energy-scale derivation itself is biased — the first thing to check then being selection bias, since the response is measured only on τ that were both reconstructed and passed ID.

What the offset is *not*. Two candidates were measured and excluded. *(i) The sampled τ_h mass.* The FastMTT floor $x_{\min} = (m_{\text{vis}}^{\text{leg}}/m_\tau)^2$ only exceeds a typical fitted $x \simeq 0.5$–0.9 once the leg mass passes ~ 1.26 GeV, so below that the assigned mass is inert: displacing the entire low-mass region by 200 MeV moves $\langle m_{\tau\tau}\rangle$ by 0.00%, against -2.4% for the same displacement in the a_1 tail. Replacing the 21-quantile grid by an explicit atom at $m_{\pi^\pm}$ plus a histogram CDF — which is both more faithful to a distribution that genuinely has an atom, and mergeable across shards —

changes $\langle m_{\tau\tau}\rangle$ by 0.11%. *(ii) A τ energy resolution mismatch.* Widening the response from 7% to 25% at fixed median moves the mean by 0.05%; it does broaden the $m_{\tau\tau}$ RMS by $\sim 2\%$, which is worth correcting for shape, but it cannot shift the peak. Only the energy *scale*, which enters $m_{\tau\tau}$ linearly, can produce a several-GeV offset.

1.5 The CMS Run-3 DNN inputs

The published CMS Run-3 analysis extracts the signal with a DNN over 26 inputs. Overlaying them is a useful audit independently of NSBI, because it makes explicit which CMS inputs Delphes cannot supply at all. Thirty quantities — p_T^{miss} (p_x, p_y) and (E, p_x, p_y, p_z) for each of ℓ_1, ℓ_2, b_1, b_2, $H_{\tau\tau}$, H_{bb} and $H_{bb\tau\tau}$, in the frame CMS uses (every momentum rotated so the visible di-τ lies at $\phi = 0$) — are built by the same script and the same selection as the NSBI features. Six inputs are not, for the reasons in Table 3.

CMS input	why it is not overlaid
decay mode (ℓ_1, ℓ_2)	Delphes has no τ substructure at all: a τ_h is an AK4 jet, so 1-prong / 1-prong+π^0 / 3-prong is undefined
charge (ℓ_1, ℓ_2)	Delphes jets carry an approximate *jet* charge, not a reconstructed τ charge
$\mathrm{cov}(p_T^{\text{miss}})_{xx,xy,yy}$	no Delphes equivalent; only an overall resolution. A synthesised covariance would be an assumption, not a measurement
ParticleNet / UParT score	Delphes gives a binary tag bit, not a continuous discriminant; our re-tag reproduces the working-point efficiency only
HHbTag score	a CMS-specific DNN over event-level inputs
pair type	we run the $\tau_h\tau_h$ channel only
FatJet block	*out of scope*, not a limitation: we run the resolved category only

Table 3: CMS Run-3 DNN inputs that Delphes cannot supply.

Four of the six — both decay modes and both charges — are precisely what a particle-flow rebuild would recover, since all four follow from resolving the prongs and π^0s inside the τ. The other two would remain missing either way. This bears directly on the decision in Section 2.

2 The decision: keep correcting the jet, or rebuild the τ_h?

Items 1 and 6 of Section 1.2, and arguably the $\Delta R_{\tau\tau}$ disagreement, all descend from representing a τ_h as a jet. The alternative is to rebuild it from the Delphes particle-flow candidates, which the card already writes (the three `EFlow` collections — tracks, photons, neutral hadrons — kept, per the card header, precisely for "τ redefinition").

What rebuilding would involve. An HPS-like construction: seed from a jet, collect charged hadrons and photons in a narrow signal cone (fixed $\Delta R \simeq 0.1$, or the p_T-dependent $3.0/p_T$ bounded at 0.05–0.1), form the standard decay-mode hypotheses (1-prong, 1-prong+π^0, 3-prong), take the τ_h four-vector as the sum of signal-cone constituents, and use an isolation annulus as a fake handle.

In favour. Mass, energy and direction become correct *by construction* rather than by correction, and three of the current corrections — the visible-mass map, the CMS-anchored energy scale and the visible-τ reference — are *retired* rather than joined by a fourth. For a phenomenological paper this is a materially stronger statement than a jet-plus-corrections chain, and it removes a fast-simulation limitation instead of documenting one. It would also recover four of the six CMS DNN inputs that Delphes currently cannot supply at all (Section 1.5), which is a sharper statement of the gain than "a better $m_{\tau\tau}$". On cost, the implementation is contained and is *not* on the critical path: running the full 5×10^5-event samples through the ladder and training the NSBI is the multi-day item.

Against, or at least uncertain. It cannot reproduce DeepTau, so the identification efficiency and fake-rate maps remain (they already work). It is *not* obviously the cure for $\Delta R_{\tau\tau}$: CMS HPS also seeds from AK4 jets, so the two merging scales are similar and a narrow signal cone may not change which pairs are resolved. And the current jet-based treatment now agrees with CMS on eight of ten features and on the yield, so the marginal gain may be small.

The caveat we most want checked. Delphes neutral particle-flow candidates are *calorimeter towers* ($\sim 0.02 \times 0.02$ in the barrel), not clustered PF objects. Since the π^0 content sets both the visible mass and the decay-mode assignment, an HPS-like reconstruction on Delphes would be approximate in exactly the place that matters most — possibly no better than the anchor-sampled mass we already use.

Questions for τ expertise.

1. Is tower-granular neutral reconstruction adequate for a useful decay-mode assignment, or does it degrade the π^0 content enough that the rebuilt mass is no better than the anchor-sampled one?
2. Signal cone: fixed 0.1 or p_T-dependent, at the p_T range that dominates here (20–60 GeV)?
3. Seeding: jets (as CMS HPS does) or direct clustering of particle-flow candidates? Would either change the efficiency for collimated pairs, $\Delta R_{\tau\tau} \lesssim 1$?
4. Is the observed pattern — over-efficient at gen $\Delta R_{\tau\tau} > 2.5$, possibly under-efficient below 1 — recognisable as a jet-versus-HPS effect at all, or should we look elsewhere?
5. Is a jet-based τ_h with anchor-measured corrections defensible in a phenomenological paper, or is rebuilding the expected standard?
6. Separately: is a 1.18× analysis-level yield excess acceptable for an unbinned NSBI study, where training uses shapes and the normalisation is fitted?

3 Reference: the studies behind these numbers

Each item is reproducible from the `delphes-pipeline` repository; scripts are named where relevant.

1. **$\tau_h\tau_h$ cutflow** (`scripts/tauh_cutflow.py`). Five stages — gen visible τ_h, reco object, ID, cleaned jets, FastMTT — walked identically on both samples, comparing *conditional* efficiencies. Initially the FastMTT stage alone carried the whole deficit (Delphes retained 12% against CMS's 99%); it now agrees at every stage, total 1.01. Unit tests inject a loss at one known stage at a time and assert it is attributed there and nowhere else.
2. **The FastMTT failure.** The hadronic decay prior is zero unless $x_{\min} = (m_{\mathrm{vis}}/m_\tau)^2 \leq 1$. A Delphes τ-jet carries the AK4 jet mass ($\mathcal{O}(5\text{–}20)$ GeV) against $m_\tau = 1.777$ GeV, so *no* solution exists and $m_{\tau\tau}$ returns NaN — silently removing ~89% of pairs. Fixed by assigning the anchor τ_h visible mass.
3. **Why the mass must be sampled, not averaged.** The visible mass is a one-sided *lower bound* on the energy fraction, not a smearing; its contribution to m_{vis} is negligible. Assigning a single median therefore *relaxes* a real constraint for every leg whose true mass is higher. Worked case: a 3-prong a_1 ($m_{\mathrm{vis}} = 1.26$, floor $x_{\min} = 0.503$) given the median 0.77

($x_{\min} = 0.188$) moves from $x = 0.525$ to $x = 0.250$, and $m_{\tau\tau}$ from 224 to 470 GeV. Over a realistic decay-mode mixture the median inflates the width ∼60% and the mean ∼9% while leaving the *mode* unchanged — so a mode-gated $Z \to \tau\tau$ candle would not catch it. The map therefore stores per-p_T quantiles and is sampled.

4. **The reference error, twice.** A Delphes `GenJet` is also a jet and also contains the underlying event, so it is *not* the analogue of `GenVisTau`. Profiled against a genuine visible τ (built as $p_{\text{vis}} = p_\tau - p_{\nu_\tau}$, valid because a hadronic decay has exactly one neutrino), the Delphes τ-jet is 17% harder at 25 GeV, falling to ∼2% at 250 GeV. The same error appeared independently in the energy-scale map and in the cutflow's stage-1 acceptance.

5. **Order of application.** The maps are all measured against CMS-scale quantities, so they must be *applied* at a CMS-scale p_T. Drawing the tags on raw jets evaluated a steeply-rising efficiency ∼17% too far right; applying the energy scale first moved the analysis-level excess $1.25 \to 1.18$.

6. **Generator-level $\Delta R_{\tau\tau}$** (`scripts/tauh_cutflow.py --gen-dr`). Two panels: the gen ΔR spectrum (do the samples *start* the same?) and the selection efficiency against it (do they *select* it the same?). Only the second is a detector statement. Result quoted in Section 1.

7. **Gen-τ construction check** (`scripts/gen_tau_check.py`). The CMS anchor carries both `GenPart` and `GenVisTau`, so our visible-τ construction can be scored against CMS's own definition on the same events: efficiency 0.991, purity 0.998, and it reproduces CMS's stage-1 rate to ∼1.6%.

8. **The m_{vis} split** (`scripts/nsbi_overlay.py --diagnostics`). The visible di-τ mass is plotted alongside $m_{\tau\tau}$ and the FastMTT inputs. Because it uses neither the p_T^{miss} nor the fit, it attributes an $m_{\tau\tau}$ offset to one of two disjoint causes. Three unit tests pin the properties the reading depends on: $m_{\text{vis}} \le m_{\tau\tau}$ always (since $x \le 1$); m_{vis} is invariant under a p_T^{miss} change ($40 \to 200$ GeV leaves it bit-identical while moving $m_{\tau\tau}$); and m_{vis} scales exactly linearly with the τ energy scale. Result in Section 1.4.

9. **Map accuracy audit.** Every tuning map was checked for statistical adequacy at the configured 2×10^5 anchor events, and none is limiting: b-tag efficiency 0.25–0.42% per bin, τ_h efficiency 0.43–1.26%, energy scale 0.2–0.3%, p_T^{miss} resolution 0.16–0.21%. What limits a map is the numerator np, not n. Two structural defects were found that more events do *not* fix — the τ_h mass quantile grid interpolates linearly across the physically empty region between the $\pi^\pm$ atom and the ρ shoulder, over-populating it by $\sim 6\times$; and every map is flat-extrapolated above 300 GeV, the last bin edge — but the first was measured to be inert (Section 1.4) and the second is untested. The second is the one worth checking, since κ_λ discrimination lives at high m_{HH} and therefore at high jet p_T.

10. **Generator copies.** Three of our own errors had one cause: Delphes writes the full Pythia history, so each physical τ appears several times, while CMS NanoAOD `GenPart` is pruned. The rule that emerged — anything that **counts** generator objects must collapse copies first; anything that **matches** them geometrically need not, since copies are collinear. This is why the energy-scale maps, the re-tagging and the overlay were never affected, while a stage-1 *count* was.

Reproducing. Derive the maps, then run the ladder and the comparison:

```
# NB --max-events is required: derive_maps caps the ANCHOR from the config but reads
# the Delphes side uncapped, so without it this reads the whole production into memory.
python -m delphes_pipeline.tuning.derive_maps --config config.v1.yml \
    --output cards/tuning/maps_v1.json --max-events 200000
```

```
python -m delphes_pipeline.validation.run_validation --config config.v1.yml
python scripts/tauh_cutflow.py --config config.v1.yml [...] --gen-dr plots/gen_dr
python scripts/nsbi_overlay.py --config config.v1.yml [...] \
    --tautau-only --diagnostics
```